

Real-World Problems and Significant Digits

Teacher's Edition

The Period

MA.8.NSO.1.6 Solve real-world problems involving operations with numbers expressed in scientific notation.

Must Solve real-world problems with numbers in scientific notation, and recognise the importance of significant digits when physical measurements are involved.

Must not The within-2 exponent limit still applies to addition and subtraction here. Significant digits limit a reported answer; they never change the arithmetic, and a count is never a measurement.

Target I can solve a real-world problem with numbers in scientific notation and report the answer with the right number of significant digits.

MTR MTR.6.1 — Boards 2, 6 and 7 · **MTR.7.1** — Every problem from Notes II on · **MTR.4.1** — Board 6

Segment	Slides	Time
Settle in; post the learning target	1	1 min
Warm-Up	2-3	5 min
Notes	4-6	12 min
Example 1	7-8	3 min
Your Turn	9-10	3 min
Example 2	11-12	3 min
Your Turn	13-14	3 min
Whiteboards	15-32	12 min
Independent Set	33	6 min
IXL	34	5 min
Total		53 min

Say these three things out loud today

1. A measurement's last digit is uncertain. That is what the whole rule is protecting.
2. Products and quotients count digits. Sums and differences count places. Two rules, and it matters which one you reach for.
3. Do all the arithmetic first and round once, at the end. Rounding in the middle loses digits the arithmetic still needed.

Slide by Slide

1. **Real-World Problems and Significant Digits.** Post the learning target.
2. **Warm-Up — On your own. Four minutes. No notes.** Q3 and Q4 are today's two tools.
3. **Warm-Up — Answers.** Reveal. 1. 1.2×10^4 2. 5.89×10^7 3. 9.02×10^{-3} 4. 4.74
4. **Notes — Copy the table.** Scientific notation makes this easy. In 5.2 times ten to the fourth, every digit you can see is significant — that is one of the reasons the notation exists.
5. **Notes.** Two significant digits in, two significant digits out. The answer is never better than the measurement it came from.
6. **Notes — Copy all three lines.** Products count digits. Sums count places. They are two different rules and it matters which one you reach for.
7. **Example 1 — A product of two measurements.** SPOKEN ONLY.
8. **Example 1 — Worked.** Multiply in full — 10,750 — and round once.
9. **Your Turn — Same steps, your numbers. Boards up when done.** Same shape, the counts the other way round.
10. **Your Turn — Answer.** Reveal; name what a wrong board most likely did (see the note above).
11. **Example 2 — A difference of two measurements.** SPOKEN ONLY.
12. **Example 2 — Worked.** Subtract in full, then stop at the leftmost last-digit place.
13. **Your Turn — Same steps, your numbers. Boards up when done.** 3,850 stops at the tens place and 274 stops at the ones, so the answer stops at the tens.
14. **Your Turn — Answer.** Reveal; name what a wrong board most likely did (see the note above).
- 15-16. **Board 1.** How many significant digits are in 0.00470 ? → **3** 2 — the trailing zero was dropped [B1G-M 8.NSO.1.6: a trailing zero after a decimal point is significant]; 5 — the leading zeros were counted [B1G-M 8.NSO.1.6: leading zeros are never significant]
- 17-18. **Board 2.** A pond's surface measures 1.4×10^2 m by 8.5×10^1 m. The pond is 3 metres deep. What is the surface area, to the right number of significant digits? → **1.2×10^4 square metres** The depth of 3 metres is not needed. anything built from the depth — the unneeded figure was used [an area needs two dimensions]; 1.19×10^4 — the arithmetic without the rounding [8.NSO.1.6 clarification]
- 19-20. **Board 3.** Two measurements are multiplied. One has 3 significant digits and the other has 5. The exact product is 7.4126×10^5 . How should the answer be written? → **7.41×10^5** 7.4×10^5 — two digits, the wrong one of the two counts [8.NSO.1.6: the answer takes the smaller count, which is 3]; 7.4126×10^5 — no rounding at all
- 21-22. **Board 4.** $(2.5 \times 10^3) + (4.71 \times 10^2)$ metres, both measured. Write the sum with the correct precision. → **3.0×10^3 m** 2.971×10^3 — the arithmetic without the precision rule [8.NSO.1.6 clarification]; 2.9×10^3 — 2,971 rounded down instead of up [rounding to the hundreds place]
- 23-24. **Board 5.** How many significant digits are in 0.03080? → **A — 4** (B) the trailing zero was dropped [B1G-M 8.NSO.1.6: a trailing zero after a decimal point is significant]; (C) one of the leading zeros was counted [B1G-M 8.NSO.1.6: leading zeros are never significant]; (D) only the nonzero digits were counted [B1G-M 8.NSO.1.6: a zero

between nonzero digits is always significant] Split (no option near two-thirds)? Sixty seconds, convince your neighbour, re-vote, then reveal.

25-26. Board 6. A rectangle measures 6.0×10^2 cm by 4.35×10^2 cm. Which is the area, to the correct significant digits? → **$A - 2.6 \times 10^5 \text{ cm}^2$** (B) the exact product, not reduced to the two significant digits the measurements carry [8.NSO.1.6 clarification]; (C) the exact product, not reduced, and 26.1 is not less than 10 — two mistakes [B1G-M 8.NSO.1.5 misconception: leaving 12×10^9 instead of 1.2×10^{10}]; (D) the exponent was not raised when the coefficient was repaired [8.NSO.1.5: point one place left, exponent up one] Split (no option near two-thirds)? Sixty seconds, convince your neighbour, re-vote, then reveal.

27-28. Board 7. A car travels 4.20×10^2 km in 3.0 hours. What is its average speed, to the right number of significant digits? → **$1.4 \times 10^2 \text{ km/h}$** 140.0 — four digits claimed from a two-digit measurement [8.NSO.1.6 clarification]; 1.4×10^1 — the exponent is one too low

29-30. Board 8. A measured length times a measured width gives an area of $3.6 \times 10^3 \text{ m}^2$. The length measured 9.0×10^1 m. What did the width measure, to two significant digits? → **$4.0 \times 10^1 \text{ m}$** 4×10^1 — one significant digit where two were asked for [8.NSO.1.6: a trailing zero after a decimal point carries the second digit]; 4.0×10^2 — the exponent is one too high

31-32. Board 9. Ellis says that because 1,234 has 4 significant digits and 5.6 has 2, the sum $1,234 + 5.6$ must be written with 2 significant digits, as 1.2×10^3 . Explain why he is wrong, and give the correct sum. → **$1,240 = 1.240 \times 10^3$** 'he used the wrong rule' — the right rule is not stated [the item asks for an explanation]; 1,239.6 — no rounding at all, which claims a tenth that 1,234 never measured

33. Independent Set — Six questions. On your own, in silence. Hand out the Independent Set.

34. IXL — Last five minutes. IXL: every listed skill is required, SmartScore 67.

Misconceptions to Watch

Board 1. 2 — the trailing zero was dropped [B1G-M 8.NSO.1.6: a trailing zero after a decimal point is significant]

Board 2. anything built from the depth — the unneeded figure was used [an area needs two dimensions]

Board 3. 7.4×10^5 — two digits, the wrong one of the two counts [8.NSO.1.6: the answer takes the smaller count, which is 3]

Board 4. 2.971×10^3 — the arithmetic without the precision rule [8.NSO.1.6 clarification]

Board 5. the trailing zero was dropped [B1G-M 8.NSO.1.6: a trailing zero after a decimal point is significant]

Board 6. the exact product, not reduced to the two significant digits the measurements carry [8.NSO.1.6 clarification]

Board 7. 140.0 — four digits claimed from a two-digit measurement [8.NSO.1.6 clarification]

Board 8. 4×10^1 — one significant digit where two were asked for [8.NSO.1.6: a trailing zero after a decimal point carries the second digit]

Board 9. 'he used the wrong rule' — the right rule is not stated [the item asks for an explanation]

Materials Whiteboards and markers. Calculators are allowed on everything and are genuinely useful today, because the point of the lesson is what to do with a long display rather than how to produce one.